

AYUSH SHARMA

Cloud Infrastructure & Distributed Systems Engineer

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SUMMARY

Software engineer with **2+ years** building and operating production **distributed systems**, **microservices**, and **event-driven Kafka pipelines** for risk, fraud, and credit platforms. Owns **8+ production services** handling **50K events/day** with **99.9% uptime**, covering monitoring, on-call incident response, performance tuning, and cloud cost optimization. Strong fit for cloud platform infrastructure across **Java**, **Python**, **Scala**, **AWS**, **GCP**, **Kubernetes**, **Kafka**, **Prometheus/Grafana**, **Redis**, and **Databricks**.

TECHNICAL SKILLS

Cloud Platform	AWS (Lambda, EC2, DynamoDB), GCP, Docker, Kubernetes, Spinnaker, CI/CD, Cloud Infrastructure, Cost Optimization, System Design, Backend Development
Backend	Java, Python, Scala, Golang, Spring Boot, RESTful APIs, Microservices, Distributed Systems, API Design, Data Structures & Algorithms, Springboot
Data & Streaming	Apache Kafka, Akka Streams, Apache Airflow, Databricks, Event-Driven Architecture, Batch/Stream Processing
Databases	PostgreSQL, DynamoDB, Redis Cloud, Elasticsearch, TimescaleDB, SQL/NoSQL, Caching
Operations	Prometheus, Grafana, Kibana, Alerting, Incident Response, RCA, On-call, SLA Ownership, Linux/OS Fundamentals

PROFESSIONAL EXPERIENCE

Software Development Engineer

June 2024 – Present

Jupiter Money — Risk Tech Team

- Own **5-6 microservices** and **3 major production services**; maintain **99.9% uptime** through on-call rotation, alerting, RCA, and deployment stewardship.
- Build cloud-backed **microservices** and **CI/CD pipelines** ingesting internal and third-party data at **10K events/day** for risk workflows.
- Integrated **3-4 third-party sources** into **event-driven fraud detection** pipelines, blocking **20% more fraudulent transactions** and reducing false positives.
- Led **Super Money**, **CIBIL**, and **PayU** integrations end-to-end; defined **API contracts**, coordinated stakeholders, and delivered production rollouts.
- Migrated **Airflow DAGs** from **Athena** to **Databricks** and cache from **AWS Elasticsearch** to **Redis Cloud**, improving platform reliability and latency.
- Upgraded **Java microservices** from **17 to 21** and tuned **AWS** resources, achieving **30% infrastructure cost reduction**.
- Revamped **Credit Risk and Fraud Detection stack** for a new business line, accelerating **risk model deployment** and production launch readiness.

Software Development Engineering Intern

Nov 2023 – June 2024

Jupiter Money — Risk Tech Team

- Built **real-time ingestion pipelines** using **Kafka** and **Akka Streams** with **DynamoDB** and **PostgreSQL**, achieving **sub-100ms latency**.
- Developed **5-6 Airflow DAGs**, **3-4 AWS Lambdas**, and **Spring Boot APIs**; reduced deployment automation time from **30 min to 5 min**.
- Implemented production monitoring with **Prometheus**, **Grafana**, and **Kibana** for critical pipelines; supported launches, incident triage, and **RCAs**.
- Recognized for early ownership of critical production components and reliable delivery across distributed backend workflows.

EDUCATION

B.Tech, Computer Science and Engineering

2020 – 2024

Indraprastha Institute of Information Technology, Delhi (IIITD) | CGPA: 8.0 / 10

KEY PROJECTS

Metrics Publisher Service [\[GitHub\]](#)

Java 17, Spring Boot, Apache Kafka, TimescaleDB, Redis, Docker, Gradle

- Engineered a **distributed, high-throughput** time-series metrics platform with a **4-stage Kafka event pipeline** and Dockerized infrastructure.
- Implemented **batch processing** (500 metrics/batch), **Redis-based cardinality control** (1K series/metric), and **T-Digest** quantile estimation.
- Exposed **REST APIs** for **P95**, **SUM**, **AVG**, **RATE** aggregations with time-bucketed queries, **90-day retention**, and **TimescaleDB compression**.

Flow-Controlled Inter-Process Communication [\[GitHub\]](#)

C, Unix Domain Sockets, FIFOs, Message Queues, Linux IPC, Make

- Implemented reliable **Linux IPC** across Unix domain sockets, FIFOs, and message queues with explicit synchronization and error handling.
- Built flow control with acknowledgement-based batching for **50 generated messages**, sending only the next batch after receiver confirmation.
- Compared transport behavior across socket, FIFO, and kernel message queue mechanisms to strengthen OS and networking fundamentals.

CERTIFICATIONS & RECOGNITION

- Best Performer of the Quarter** — Jupiter Money (Awarded for consistent delivery and production ownership)
- ARTPARK Fellowship** — IISc Bangalore (Awarded for research contributions)
- Amazon ML Summer School** — Amazon (Competitively selected), 2022

ADDITIONAL

- Teaching Assistant for Basic Electronics (300+ undergraduates) at IIITD
- Director, Communication & Marketing at TEDxIIITD